

MATH 231 – Linear Algebra I

The Vector Unit, Part 4 ■ The Standard Basis, Span, and Coordinates

Kiely Hall 326 ■ Instructor: Kevin Bedoya

About this document. This is Part 4 of the vector unit. Section numbering runs continuously through all parts, so a reference such as §12.4 points into Part 2 and means exactly what it says:

Part	Sections	Subject
1	§1–§9	vectors, notation, addition, scaling, equality
2	§10–§12	orientation, the dot product, projection
3	§13–§15	Cauchy–Schwarz, metrics, norms
4	§16–	the standard basis, span, coordinates

Assumed from Parts 1–3. Addition and scalar multiplication (§7–§8); the bracket notation $\mathbf{v} = [v_1, v_2]$ and $\mathbb{R}^2$ (§6); the orthogonality test $\langle \mathbf{u}, \mathbf{v} \rangle = 0$ (§11.7); the projection $\text{proj}_{\mathbf{u}}(\mathbf{v})$ and component $\text{comp}_{\mathbf{u}}(\mathbf{v})$ (§12.4); and what it means for two vectors to be parallel (§10.1).

What you should be able to do afterwards. Write down the standard basis vectors and verify that they are orthonormal; express any vector of $\mathbb{R}^2$ as a linear combination of them; say what *span* and *basis* mean; explain what the coordinates of a vector are and why they are unique; set up the problem of finding coordinates as a linear system; say what the dimension of a space is and why it agrees with the degrees-of-freedom count of §1; produce bases other than the standard one; recognise when two vectors fail to span the plane; and define linear independence and dependence.

16. The standard basis of $\mathbb{R}^2$

Since §6 we have written a vector as a bracketed pair, $\mathbf{v} = [v_1, v_2]$, and treated those two numbers as simply given. This section asks where they come from. The answer is that the bracket notation has been making a silent choice all along — and once we see what the choice is, we can start asking what else could have been chosen.

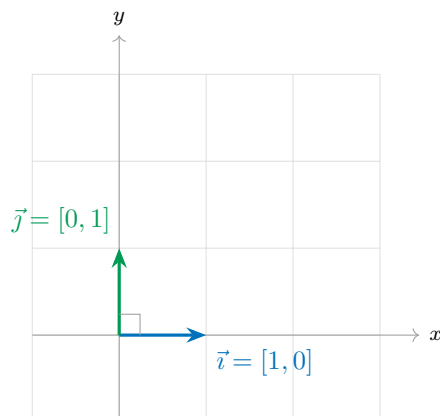
16.1. Two special vectors

Definition 16.1 (The standard basis vectors). The **standard basis vectors** of $\mathbb{R}^2$ are

$$\vec{i} = [1, 0], \quad \vec{j} = [0, 1].$$

They are written with arrows overhead, one of the three conventions of §6.

Geometrically $\vec{i}$ is one step east and $\vec{j}$ is one step north: the two elementary moves, one along each axis.



Three properties, each of which we already have the tools to check.

- **Both have unit length.** $\|\vec{i}\| = \sqrt{1^2 + 0^2} = 1$ and $\|\vec{j}\| = \sqrt{0^2 + 1^2} = 1$, so both lie on the unit circle $\mathbb{S}^1$ of §12.1.
- **They are orthogonal.** Their dot product is $\langle \vec{i}, \vec{j} \rangle = 1 \cdot 0 + 0 \cdot 1 = 0$, so by the orthogonality test of §11.7 they meet at a right angle — as the figure shows, since they lie along the two axes.
- **Neither is a multiple of the other.** No scalar α satisfies $\alpha[1, 0] = [0, 1]$, since the first coordinate would force $\alpha = 0$ and the second would force $\alpha \cdot 0 = 1$. So by §10.1 they are not parallel. Neither one is redundant: you cannot manufacture $\vec{j}$ out of $\vec{i}$ alone.

A collection of vectors that are mutually orthogonal and each of unit length is called **orthonormal** — “ortho” for the right angles, “normal” for the unit lengths. So $\vec{i}$ and $\vec{j}$ form an orthonormal pair, which is as convenient an arrangement as two vectors can have. §16.4 shows what that convenience buys.

A common confusion. This course uses the letters i , j and k for three different things, and you will meet all of them. In the complex numbers lecture, i is the *imaginary unit* with $i^2 = -1$, and i, j, k are the *basis elements* of the quaternions. Here $\vec{i}$ and $\vec{j}$ are *vectors* in $\mathbb{R}^2$. The arrows are what tell them apart, so keep them on.

16.2. Building vectors out of them

Take any two real numbers α and β and form

$$\alpha\vec{i} + \beta\vec{j}.$$

Nothing here is new: it is scalar multiplication from §8 followed by addition from §7. Working it out,

$$\alpha\vec{i} + \beta\vec{j} = \alpha[1, 0] + \beta[0, 1] = [\alpha, 0] + [0, \beta] = [\alpha, \beta],$$

which is a vector in $\mathbb{R}^2$. So every such combination lands inside $\mathbb{R}^2$.

Definition 16.2 (Linear combination). An expression of the form $\alpha\mathbf{u} + \beta\mathbf{v}$, with $\alpha, \beta \in \mathbb{R}$, is called a **linear combination** of $\mathbf{u}$ and $\mathbf{v}$.

The interesting direction is the reverse one. Does every vector of $\mathbb{R}^2$ arise this way, or do these combinations only reach part of the plane? They reach all of it, and the proof is a single line: given any $\mathbf{v} = [v_1, v_2]$, choose $\alpha = v_1$ and $\beta = v_2$, and

$$v_1\vec{i} + v_2\vec{j} = [v_1, v_2] = \mathbf{v}.$$

Nothing in $\mathbb{R}^2$ is missed. Two vectors, scaled and added, generate the entire plane — which is why $\vec{i}$ and $\vec{j}$ deserve to be called the *scaffolding* of $\mathbb{R}^2$: build them once and everything else follows.

16.3. Span and basis

Those two paragraphs proved two inclusions, and together they deserve a name.

Definition 16.3 (Span). The **span** of a collection of vectors is the set of *all* their linear combinations. For $\vec{i}$ and $\vec{j}$,

$$\text{span}\{\vec{i}, \vec{j}\} = \{ \alpha\vec{i} + \beta\vec{j} \mid \alpha, \beta \in \mathbb{R} \}.$$

The first calculation of §16.2 showed $\text{span}\{\vec{i}, \vec{j}\} \subseteq \mathbb{R}^2$, and the second showed $\mathbb{R}^2 \subseteq \text{span}\{\vec{i}, \vec{j}\}$. Both inclusions hold, so

$$\boxed{\text{span}\{\vec{i}, \vec{j}\} = \mathbb{R}^2}$$

and we say that $\vec{i}$ and $\vec{j}$ **span** $\mathbb{R}^2$.

Definition 16.4 (Basis, working definition). A collection of vectors is a **basis** for a space when it spans that space and does so without redundancy — no vector in the collection is a linear combination of the others, so none of them can be dropped. We write

$$\mathcal{B} = \{ \vec{i}, \vec{j} \},$$

and call $\mathcal{B}$ the **standard basis** of $\mathbb{R}^2$.

Remark (Basis and span are different objects). Keep the two apart. The basis $\mathcal{B} = \{ \vec{i}, \vec{j} \}$ is a set containing *two vectors*. Its span, $\text{span}\mathcal{B} = \mathbb{R}^2$, is a set containing *infinitely many* — every vector in the plane. A basis is small; what it generates is large. That is the entire point of a basis: it is a finite description of an infinite set.

Redundancy is the clause worth dwelling on. Spanning alone is easy to achieve by piling on extra vectors: the collection $\{ \vec{i}, \vec{j}, [3, 5] \}$ also spans $\mathbb{R}^2$, but the third vector earns nothing, since $[3, 5] = 3\vec{i} + 5\vec{j}$ was already reachable. A basis is required to carry no such passengers, and §16.1 checked that ours does not: neither $\vec{i}$ nor $\vec{j}$ is a multiple of the other.

Looking ahead. “Without redundancy” is doing real work here and deserves a sharper statement than the one above. It has a proper name — *linear independence* — and a definition that applies to any number of vectors at once, not just two. We will meet it shortly.

16.4. Coordinates

Definition 16.5 (Coordinates). If $\mathbf{v} = \alpha\vec{i} + \beta\vec{j}$, the scalars α and β are the **coordinates** of $\mathbf{v}$ with respect to the basis $\mathcal{B}$.

For the word “the” to be justified, there had better be only one such pair — and there is. Suppose a vector could be written two ways,

$$\alpha\vec{i} + \beta\vec{j} = \alpha'\vec{i} + \beta'\vec{j}.$$

Both sides evaluate componentwise to $[\alpha, \beta]$ and $[\alpha', \beta']$, and two vectors are equal exactly when their coordinates agree (§9). So $\alpha = \alpha'$ and $\beta = \beta'$: the two representations were the same one. Coordinates are **unique**.

Remark (What the bracket notation was hiding). Look again at $\mathbf{v} = [v_1, v_2]$. We have just shown $\mathbf{v} = v_1\vec{i} + v_2\vec{j}$, so the two numbers inside the brackets are exactly the coordinates of $\mathbf{v}$ with respect to $\mathcal{B}$. The notation of §6 was never neutral — it silently reported coordinates in one particular basis. We have been able to ignore this only because $\mathcal{B}$ has been the sole basis in play. Other bases exist, and the same vector has different coordinates in each.

Remark (Why reading them off is free). There is a reason the coordinates turned out to be the components themselves. Because $\mathcal{B}$ is orthonormal, the projection machinery of §12.4 collapses:

$$\text{comp}_{\vec{i}}(\mathbf{v}) = \frac{\langle \vec{i}, \mathbf{v} \rangle}{\|\vec{i}\|} = \frac{1 \cdot v_1 + 0 \cdot v_2}{1} = v_1,$$

and likewise $\text{comp}_{\vec{j}}(\mathbf{v}) = v_2$. So the coordinate of $\mathbf{v}$ along a basis vector *is* its projection component onto that basis vector — the shadow it casts on each axis. Orthonormality is what makes the norms 1 and the cross terms 0, and so makes the arithmetic vanish.

16.5. The reverse problem, and a first linear system

Turn the question around. Instead of being handed α and β and asked what they build, start with a vector $\mathbf{v} \in \mathbb{R}^2$ and treat the scalars as *unknowns*:

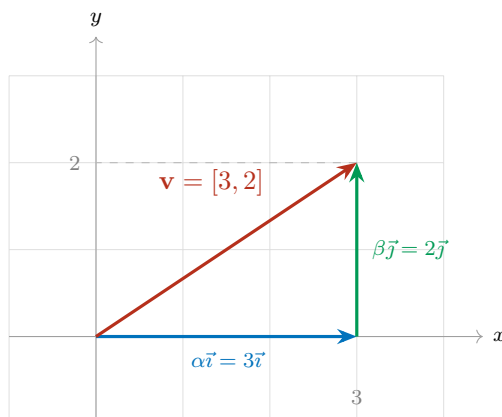
$$\mathbf{v} = \alpha\vec{i} + \beta\vec{j}, \quad \alpha, \beta \text{ to be found.}$$

Two facts already settle that this is a sensible question. Because $\vec{i}$ and $\vec{j}$ span $\mathbb{R}^2$, a solution *exists* — no vector of the plane is out of reach. Because coordinates are unique, the solution is the *only* one. We are not searching hopefully; we are computing something we know is there and know is unambiguous.

The picture

Geometrically the answer is a walk. Starting at the origin, travel α steps along $\vec{i}$ — purely horizontal — and then, from where you land, β steps along $\vec{j}$ — purely vertical. That is the head-to-tail reading of addition from §7.1, and the question is how far to walk in each direction to arrive exactly at the head of $\mathbf{v}$.

Read the other way, the dashed guides drop from the head of $\mathbf{v}$ onto the two axes, and where they land is the answer.



Writing it as a system

Now do it algebraically. Expand the right-hand side componentwise:

$$\alpha\vec{i} + \beta\vec{j} = [\alpha \cdot 1 + \beta \cdot 0, \alpha \cdot 0 + \beta \cdot 1].$$

Setting this equal to $\mathbf{v} = [v_1, v_2]$ and matching coordinates gives two equations in the two unknowns:

$$v_1 = \alpha \cdot 1 + \beta \cdot 0,$$

$$v_2 = \alpha \cdot 0 + \beta \cdot 1.$$

A collection of equations like this — several linear equations sharing the same unknowns, to be satisfied simultaneously — is called a **linear system**. This one barely deserves the name, because the zeros knock out one term in each equation and it collapses on sight:

$$v_1 = \alpha, \quad v_2 = \beta.$$

Example 16.1. For $\mathbf{v} = [3, 2]$ the system gives $\alpha = 3$ and $\beta = 2$, so $\mathbf{v} = 3\vec{i} + 2\vec{j}$ — the walk in the figure above. For $\mathbf{v} = [-2, 5]$ it gives $\mathbf{v} = -2\vec{i} + 5\vec{j}$: two steps west, then five north.

Remark. Do not mistake how easy that was for how easy the question is. The system collapsed because the coefficient of β in the first equation and of α in the second were both 0, and those zeros are precisely the statement $\langle \vec{i}, \vec{j} \rangle = 0$ — the orthogonality we checked in §16.1, reappearing as the arithmetic that makes the work disappear. The standard basis is engineered so that reading coordinates costs nothing.

Looking ahead. Ask the same question against a basis that is *not* the standard one and the zeros are gone. The two equations then genuinely interact, both unknowns appear in both, and the system must actually be solved. That is the general problem of linear systems — how to state it, when it has a solution, how many solutions there can be, and how to compute them — and it will occupy a substantial part of this course. The trivial case above is worth remembering as the one you are always trying to get back to.

17. Dimension

A basis carries a number with it, and that number turns out to be one we have been using informally since §1.

Definition 17.1 (Dimension). The **dimension** of a space is the number of vectors in a basis for it.

Our basis $\mathcal{B} = \{\vec{i}, \vec{j}\}$ contains two vectors, so

$$\dim \mathbb{R}^2 = 2.$$

The superscript in $\mathbb{R}^2$ was telling us the dimension all along.

Why this agrees with what we already meant

In §1 we described dimension as a *degree of freedom* with respect to spatial coordinates: how many independent numbers must you supply to pin down a position? The two accounts are the same account. Supplying a position in $\mathbb{R}^2$ means supplying its two coordinates α and β — one per basis vector — so the count of free numbers and the count of basis vectors are forced to agree.

Read geometrically, the dimension counts **independent directions**. The basis $\mathcal{B}$ offers two: along the x -axis and along the y -axis. Neither can be obtained from the other, and once both are available, nothing further is needed.

A common confusion. “Two independent directions” does not mean the basis represents only two directions. It represents *every* direction in the plane — the vector $3\vec{i} + 2\vec{j}$ points in a direction that is neither of the two axes. What the number 2 counts is how many directions you must be *given* before all the rest come free. Two is the number of directions that are independent, not the number that are reachable.

18. Other bases, and linear independence

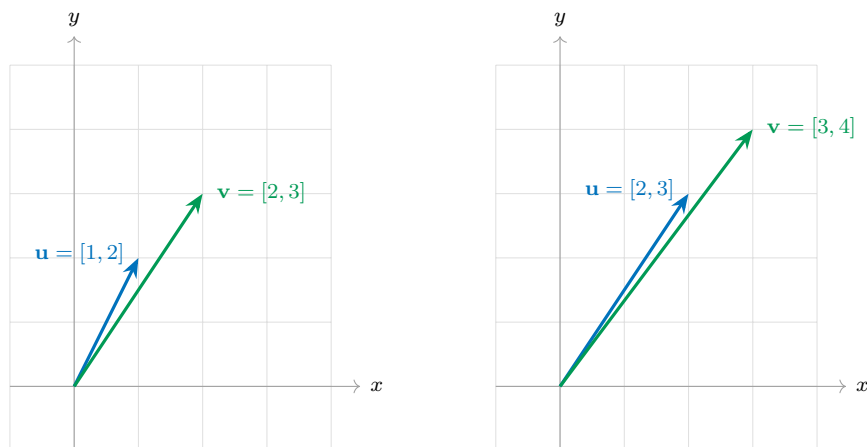
Nothing in §16.3 said the standard basis is the *only* basis for $\mathbb{R}^2$. It is merely the most convenient one. Once we go looking for others, a question appears that the standard basis was too well behaved to raise: which pairs of vectors work, and which fail?

18.1. Bases other than the standard one

Consider the pairs

$$\mathbf{u} = [1, 2], \mathbf{v} = [2, 3] \quad \text{and} \quad \mathbf{u} = [2, 3], \mathbf{v} = [3, 4].$$

Neither pair looks anything like $\{\vec{i}, \vec{j}\}$ — the vectors are not unit length, not orthogonal, not aligned with the axes — and yet each pair is a perfectly good basis for $\mathbb{R}^2$.



Being a basis costs something, though: the coordinates are no longer free to read off.

Example 18.1 (Coordinates against a non-standard basis). Take the basis $\{\mathbf{u}, \mathbf{v}\}$ with $\mathbf{u} = [1, 2]$ and $\mathbf{v} = [2, 3]$, and ask for the coordinates of $\mathbf{w} = [1, 0]$. We need α, β with $\alpha\mathbf{u} + \beta\mathbf{v} = \mathbf{w}$, which componentwise is

$$\begin{aligned} \alpha \cdot 1 + \beta \cdot 2 &= 1, \\ \alpha \cdot 2 + \beta \cdot 3 &= 0. \end{aligned}$$

No zeros this time, so nothing collapses. The second equation gives $\alpha = -\frac{3}{2}\beta$; substituting into the first, $-\frac{3}{2}\beta + 2\beta = \frac{1}{2}\beta = 1$, so $\beta = 2$ and $\alpha = -3$. Checking:

$$-3[1, 2] + 2[2, 3] = [-3, -6] + [4, 6] = [1, 0]. \quad \checkmark$$

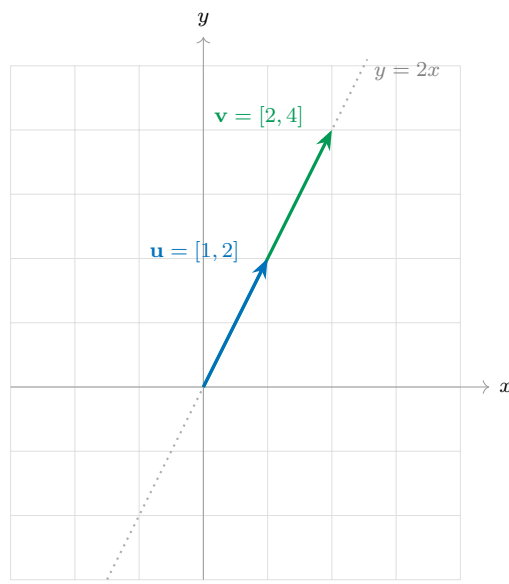
So $\mathbf{w}$ has coordinates -3 and 2 in this basis, where in the standard basis it had coordinates 1 and 0 . Same vector, different numbers — exactly what the remark in §16.4 warned. And this is the situation §16.5 pointed at: away from the standard basis, finding coordinates is a genuine linear system that has to be solved.

18.2. When two vectors do not span the plane

Now try

$$\mathbf{u} = [1, 2], \quad \mathbf{v} = [2, 4].$$

Plot them and something is visibly wrong: both arrows lie along one and the same line through the origin.



The reason is immediate: $\mathbf{v} = 2\mathbf{u}$. And that single fact collapses every linear combination onto the same line:

$$\alpha\mathbf{u} + \beta\mathbf{v} = \alpha\mathbf{u} + \beta(2\mathbf{u}) = (\alpha + 2\beta)\mathbf{u},$$

which is a scalar multiple of $\mathbf{u}$ — and by §8 every scalar multiple of $\mathbf{u}$ lies on the line through the origin and the head of $\mathbf{u}$. Whatever α and β you choose, you never leave that line.

So there is no way to assign coordinates to a point off the line: the point is simply not reachable. These two vectors span only a *subset* of the plane, and therefore

$$\text{span}\{\mathbf{u}, \mathbf{v}\} \neq \mathbb{R}^2 \quad \text{when} \quad \mathbf{u} = [1, 2], \quad \mathbf{v} = [2, 4].$$

Two vectors are not automatically a basis for $\mathbb{R}^2$. Having the right *number* of vectors is not enough; they have to stand in the right relationship to one another.

18.3. Three questions

That failure is worth more than the successes, because it tells us there is something to characterise. Three questions follow, and we take them one at a time.

Q1. Which pairs of vectors form a basis for $\mathbb{R}^2$ — that is, which pairs span $\mathbb{R}^2$?

Q2. The pair $[1, 2]$ and $[2, 4]$ failed to span $\mathbb{R}^2$, but they clearly span *something*. What kind of basis do they form?

Q3. How many different bases does $\mathbb{R}^2$ have?

18.4. Question 1: linear independence

Q1 is better asked as a question about a *relationship*: what must be true of two vectors, with respect to each other, for them to form a basis?

The two examples answer it geometrically. The vectors $[2, 3]$ and $[3, 4]$ point in genuinely different directions, and no amount of stretching turns one into the other: there is no scalar α with $\alpha[2, 3] = [3, 4]$, since the first coordinate demands $\alpha = \frac{3}{2}$ and the second then gives $\frac{3}{2} \cdot 3 = \frac{9}{2} \neq 4$. The pair $[1, 2]$ and $[2, 4]$ fails exactly here: $2 \cdot [1, 2] = [2, 4]$, so one *is* a stretch of the other, and that is what flattened their span onto a line.

So the property we want is “neither vector is a scalar multiple of the other” — which is the *not parallel* of §10.1. That phrasing is correct, but it is awkward: it treats the two vectors asymmetrically, and it does not generalise past two. The standard formulation fixes both problems by asking instead about combinations that produce the zero vector:

$$\alpha[2, 3] + \beta[3, 4] = \mathbf{0} \quad \Longleftrightarrow \quad \alpha = \beta = 0.$$

Read the left-to-right direction as the real content: the *only* way to combine these two vectors and land back at the origin is to take nothing of either.

Definition 18.1 (Linear independence and dependence in $\mathbb{R}^2$). Two vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$ are **linearly independent** when the only scalars $\alpha, \beta \in \mathbb{R}$ satisfying

$$\alpha\mathbf{u} + \beta\mathbf{v} = \mathbf{0}$$

are $\alpha = 0$ and $\beta = 0$.

They are **linearly dependent** when they are not — that is, when there exist scalars α, β , *not both zero*, with $\alpha\mathbf{u} + \beta\mathbf{v} = \mathbf{0}$.

Remark (Why “not both zero” cannot be dropped). The choice $\alpha = \beta = 0$ always works, for any two vectors whatsoever, since $0\mathbf{u} + 0\mathbf{v} = \mathbf{0}$ is automatic. It carries no information. So a definition of dependence reading “there exist α, β with $\alpha\mathbf{u} + \beta\mathbf{v} = \mathbf{0}$ ” would be satisfied by every pair and would say nothing at all. The clause *not both zero* is what gives the definition teeth — and for the same reason, independence is stated as $\alpha = \beta = 0$ being the *only* solution.

Example 18.2 (The two pairs, checked). For $\mathbf{u} = [1, 2]$ and $\mathbf{v} = [2, 4]$, take $\alpha = 2$ and $\beta = -1$:

$$2[1, 2] + (-1)[2, 4] = [2, 4] - [2, 4] = \mathbf{0},$$

and those scalars are not both zero. So the pair is linearly **dependent** — and we have already seen the consequence: their span is a line, not the plane.

For $\mathbf{u} = [2, 3]$ and $\mathbf{v} = [3, 4]$, suppose $\alpha[2, 3] + \beta[3, 4] = \mathbf{0}$. Componentwise that is $2\alpha + 3\beta = 0$ and $3\alpha + 4\beta = 0$. The first gives $\alpha = -\frac{3}{2}\beta$; substituting into the second, $-\frac{9}{2}\beta + 4\beta = -\frac{1}{2}\beta = 0$, so $\beta = 0$ and then $\alpha = 0$. The only solution is the trivial one, so the pair is linearly **independent**.

Remark (For two vectors, dependence is exactly parallelism). When $\mathbf{u}$ and $\mathbf{v}$ are both nonzero, linear dependence says nothing more than what §10.1 called parallel. If $\alpha\mathbf{u} + \beta\mathbf{v} = \mathbf{0}$ with, say,

$\alpha \neq 0$, then $\mathbf{u} = -\frac{\beta}{\alpha}\mathbf{v}$, so each is a scalar multiple of the other; and conversely a multiple relation rearranges into such a combination. The one wrinkle is the zero vector: $\{\mathbf{0}, \mathbf{v}\}$ is always dependent, since $1 \cdot \mathbf{0} + 0 \cdot \mathbf{v} = \mathbf{0}$ uses a nonzero scalar — yet §10.1 deliberately excluded $\mathbf{0}$ from the word “parallel” because it has no direction. The combination formulation handles that case without special pleading, which is one more reason it is the definition of record.

This answers Q1, at least for pairs: **two vectors form a basis for $\mathbb{R}^2$ precisely when they are linearly independent**. Independence is what guarantees the two arrows genuinely open the plane out, instead of collapsing it onto a line.

Looking ahead. The definition above was written for two vectors so that it can be checked by hand, but nothing in it depends on there being two. The general version asks the same question of any collection — when is the zero vector reachable only by taking nothing of everything? — and that is the version we will need once we leave the plane. Q2 and Q3 remain open.